

ActInf Textbook Group ~ Cohort 2 ~ Meeting 2

(Chapter 1)

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so okay hello everyone thanks for joining this second meeting of the second cohort of the active inference textbook group where we're going to be discussing chapter one today is september 9th 2022 so i'll minimize this and you guys can click on my screen and that will show you the full screen of what i am presenting so what we'll do in the live meetings here for the next two and a half months or so is totally going to be driven by the material that's in the chapters and in the appendices so we're going to go to the questions page and we're going to look at the questions that have been asked for that week for that theme and we're just going to start with the most upvoted questions so here you can upload a question like this by clicking thumbs up um so we will start with the most upvoted questions and then we can take notes here and the discourse in this answers in this course and if you just click on that it opens this page and anybody is free to write here and after addressing the questions which is probably going to be most of the time and discussing them then we will move on to maybe look at the ideas more broadly over here so we don't really have too many questions if anybody wants to jump in and write a question we did send an email reminder that um because this was pretty empty over the course of the week but hopefully thinking by writing in this way will be a structure where we'll be able to hear everyone's perspective and leave the coda in a better place than we found it so if anybody wants to jump in and write additional questions in the coda now would be a good time and then after looking at the questions we will pull back and focus a little bit more broadly so maybe we'll go over here to touch on the ideas and insights of the chapter and then ideas for that week and then in the last minutes we can just see if there's been any changes to the project ideas and start to see what people are excited about applying active inference to um and then we will end at the top of the hour so let's just head over to the questions the cohort 2 questions page and take a couple minutes to put some questions in here um so just so you guys know control plus and control minus are good ways to scroll in and out to get like the right view of the screen and then you can hide this little textbook bar over here

so i'm going to hide that and just give everybody maybe like five minutes to put in a question or something they would maybe like to discuss from chapter one of the textbook [Music] foreign so maybe we could just take like one more minute and wrap up these questions and just want to say like to everyone at any point that there's always the affordance of just listening and also of just re-watching this recording there's also the affordance in gather to raise your hand um see if i can show where that is um boy it's been a long time over here there it is down at the bottom so there's this raised hand feature and lower hand yeah okay let's see and then if you would you can raise your hand if you would like to share just on that um particular question and we're going to have them up in the coda and you can also pull up the answers um in the discourse and that will pull up the focused view of the question and everybody um is free to collaboratively like edit the answers here in the discourse it's better to really add and be noisy even if it's just like a bullet point list or an incomplete thought or if there's incoherence or if it's contradictory those are all fine it just adds information and perspective on the question um so yeah we're going to open up each question the answers in the discourse and read what has been read and already contributed in the answers and then also people want to share other thoughts that they're having that's awesome and then if anybody also just wants to be taking notes just things that are coming to mind or they don't want to share in a speaking way or just taking some notes on what other people are saying so this is really a participatory uh group and so we it's just is what we make it here okay cool i actually this little window opener i won't see any raised hands and gather all right let's see i need a double screen all right let's see if i can do it in this small way okay cool um all right so let's get into it this first question i'll pop it open and it says what is the difference between a stimulus response mapping and a state action policy so i don't know does anybody have any ideas here i'm not i think that they're both um kind of like an if this then that but they're specific i think to machine learning so if anybody has um any of like more depth or background in this area that would be awesome to hear so i don't know the way i kind of think about it oh is that uh arun did you yeah can you hear me yeah cool i can give a very i can give an attempt at an answer to that which from my understanding and please tell me if i'm wrong um when you've got the stimulus response mapping you've you're con you've got 100 certainty on what the stimulus is so with the states you're then factoring in beliefs about you've got uncertainty so maybe you just you know at a simple form you could be chucking a gaussian distribution on every stimulus and then a gaussian distribution on what your response is to go from state to go from stimulus response mapping to state action policy that's my intuitive understanding of what's going on but again

please tell me if i'm barking on the wrong track yeah that's awesome thanks for um sharing and i kind of think about it in the same way like i i don't know i mean this is probably totally wrong but like i think like about the roomba like the little vacuum that cleans your house um and so like if you turn it on it turns on and moves like that's a stimulus like you push the button and then it turns on and then um like it moves around because you pushed the power button like it's a very clear thing versus like the state action like if it's it's full like if the vacuum gets full then it like dumps out but like that state of fullness is like a right like a distribution right like so it can be like more or less full and it senses how far it is away from the dump out bucket or whatever so that's like the state action like the state can be full but like what like what is too full for the roomba like is it a certain weight that's in there or like a certain volume or sensor i don't know i have i have uncertainty about when the room dumps itself out cool awesome so we can just write this in here so um and anyone's free to take notes and type in here okay cool all right so um yeah and anyone's free to just get in here and jump in and and do this all right so let's look uh let's look at this one all right so this question says is the brain trying to maximize qualia divergence and minimize encoding length ali sorry i raised my hand before um i was just wanted to add some comments about the previous question uh because i think it's a very important one um you see uh as we go on uh specifically specifically in the chapter four when we encounter the concept of markup blanket uh will i think we can see uh kind of more uh granular differences between those two concepts between uh stimulus response mapping and state action policy but briefly states are just representations of um internal or external world right and actions are specifically uh goal-directed called goal-oriented responses and they just try to change the state of the system or the agent but when we're talking about uh just uh stimulus response mapping i mean response uh doesn't necessarily tries to change the state of the agent uh as a as an example uh if you touch a hot object the pain you feel is just a response right but if you move away your hand from that hot object it's an action which tries to change the stimulus and one other important distinction here is the term policy is used in a different sense in active inference literature as compared to reinforcement learning literature namely in reinforcement learning this term is often used to refer to singular mappings or one-to-one mappings from states to actions but in active inference literature policy usually refers to possible sequence of actions not just a singular action so that's i think one important distinction we need to have in mind also cool awesome thank you very much and i looked that up actually so here like i think that um when they talked about state action policies they were using it in the machine learning case because it

says in the textbook the goal directed character of action and active inference is in keeping with early cybernetic formulations but distinct from most current theories that explain behavior oh sorry someone's talking could you please mute thank you so it says distinct from most current theories that explain behavior in terms of stimulus response mappings or state action policies so i think that this here um they're not talking about um the act of inference state action policies but they're using it in the reinforcement learning way so it says also stimulus response or habitual behavior then comes a special case of a broader family of policies in active inference so i think that the the way that they were talking about it in the textbook refers to the reinforcement learning way as opposed to the active inference way which only is correct it's definitely um different any other comments or thoughts on this and thank you for whoever is taking notes in the coda that's always super helpful so we can move on and look at this question is the brain trying to maximize qualia divergence and minimize encoding length who has some thoughts here just um the idea of maximizing qualia divergence seems like at some point that's beyond schizophrenic maybe um so there's like some maybe thresholded which you would not want to go beyond and that has seems to have some to me seems to have something to do with epistemic value you can't just maximize try to try to maximize your epistemic value forever you would into a pragmatic problem that's encoding length so i'm just i guess i'm wondering if uh maybe a a different framing of the question or commensurate framing of the question is what's the relationship between epistemic value and pragmatic value like are there trade-offs there yeah i scanned through um chapter one and really didn't see a quality of divergence or encoding length there in the chapter so i'm not really um does is this anyone's question do you want to unpack this question a little bit more for us maybe no volunteers awesome daniel thanks for joining us i will i can officially uh pass the baton off to you and maybe you have some comments about maximizing qualia divergence or minimizing encoding length here but we got in and we wrote a few questions and we went through the first one uh we could also go back and touch on that if you've got comments on on that as well questions just makes me wonder what is being minimized what is being maximized and when is it some particular value that's being minimized or maximized and when is it the divergence between which other um differences and in chapter one there aren't any of the formalisms yet we'll see a lot of double lines and kl divergences that will cue us towards where a divergence is being at least described if you're ready i can unshare my screen and you can take over but please please continue any more additional additional comments i like this is oh yeah i thought didn't get to write this down but brock had mentioned

qualia schizophrenia beyond schizophrenia that was the term diversity you can imagine is necessary and some more might be really healthy and useful but at some point i don't then you're you have multiple personalities or something i don't that makes me think of um the sense of normalcy and what is experienced as normal and how much of a how much of a deviation from normal is perceived as like what what what is perceived as normal and where is it perceived as a creative and maybe even enjoyable novelty like a guitar solo and then where is the quality of divergence too high and maybe uncomfortable and then what's the view from the inside with that actual experience like what qualia brings in and then where where is this being modeled from the outside with behavioral descriptions where we we might not be able to directly address what the qualia are and daniel that reminds me like some of it might be you said some of it might be normal and some of it might be uncomfortable and or some of it might be some divergence might be creative and some might be uncomfortable and there's actually like a huge degree of um of like mental disorder associated with creativity like i mean studies have been done and it's like i mean you like i mean there's the classic like you know songwriter singer who commits suicide and we hear this again and again but there's actually like a great degree of mapping between the creative type and some kind of atypical mental state so maybe both of those are the the same okay cool anybody have any comments i don't know if i can see everybody in my tiny screen go ahead sorry hello hi it's not clear to me what real what they precisely means equality diver divergence because divergence with respect to what i mean usually a kl divergence is a divergence of um like a measure of the difference between two distribution so what does exactly mean maybe the person asking the question and clarify i'll also just know that they'll go bro sorry i was just i don't i don't know if they're here because we already asked but to me it just seems like uh internal kind of population of one qualia set another qualia set that are differentiable like you can have divergence certainly of qualia uh in your own experience if you just consider you know the various sensory kind of experiential states that you you occupy like okay so that means that would mean um like expanding the range of uh of your phenomenology something like that obviously yeah so um yeah so you could imagine a sense of space and self you know that's kind of uh frequent and normative that if that starts to maybe in a religious or spiritual setting or other kind of group team settings it's you know you kind of get a group flow going maybe like that that starts to change but but maybe if those two things greatly diverge that wouldn't be great so that maybe is related to the complexity of that part of the model that encodes um a representation of the self so that would increase the complexity of the of the model which is one part one part of of

the free energy minimization scheme so the i guess i mean with each increase in the epistemic uh import of uh of the of observation then also complexity of the model though needs to be kept at bay within within some limit inside your blanket so it yeah so it's probably it's a trade-off between increasing the complexity of the model for what concerns the range of possible experiential and their epistemic value i guess also just mention out of uh fairness that qualia is not mentioned in the textbook but it's totally awesome to consider what the formalisms entail for philosophy and many papers and many discussions do and it's like an important area to explore but it's also not what is um addressed and i i think it's probably even though these authors in other settings tackle the problem head on is seemingly crafted to enable some discussions that might not require the consideration of the view from the inside or the phenomenology that qualia entail and that helps the generality and the applications across different systems and so understanding how they map onto our undeniable experience is a great area to explore no no i was saying that overall the book doesn't doesn't expand too much on the on the part of the model that represents the the agent itself as a you know as an entity like you know our own self-representation it it mentions it but does not really does not really dwell onto it more about that is probably in the book of uh jacob hoey there's a little bit more on that but in this one not so much well perhaps this connects to the um following question on self-evidencing so before we jump forward um i like i just want to stay on this question for one second so i i looked up the definition of qualia here um and it's the the individual instances of subjective conscious experience and i don't know maybe in the question which let's read it again it says is the brain trying to so i was wondering if this maybe is meant for a question like in an evolutionary sense like is this what the the brain is trying to do um across time like in deep time from you know prehistory to now um yeah we can move on though because maybe um self-evidencing is is uh more addressed in the book than qualia divergence so what does self-evidence and inactivism really mean does anyone want to take a stab here yeah i ask the question because um when i read up my eyes just glaze over when i i hit these words i i can read them in sentences and i just just doesn't stick in my my head what what is really meant self-evidencing just it just turns into something tautological i don't have any sort of real grasp but what is what is being said in these sentences i can take a step at that because really it's a it's a it's quite a simple concept so if we think about the agent not as having a model of the world but being itself a model of its own environmental niche then minimizing free energy basically means uh working acting so as to collect the the observation that it expects that is minimizing surprise so since you can say finding the evidence for its model of

the world but since the agent is a model of the world that becomes self-evidencing so basically you could also see it as trying to find the sensory stimulation that is um that makes your phenotype your organism viable that sustains you so that's that's evidencing me finding the observation the sensory evidence that is more coherent with your makeup as an organism thanks i'll add a few notes so an activism is a long-lived thread in cognitive sciences that is long existing before active inference and it is describing that systems with bodies real particular things existing in the world ought to be um having their cognition modeled as bodies engaging with the world rather than like them taking like a view from the outside or um necessarily needing everything to be like symbolic communication whereas in digital systems or kind of abstractions of cognitive systems it's very easy to get into some information and symbol passing discussions whereas for real systems with biomechanics and embodiment inactivism brought attention to the requirement of considering like the active engagement of an entity with its niche maybe it's not um a term that provides a huge cognitive model update because for many people it's very natural to consider cognitive systems as inactive and embodied embedded and cultured et cetera et cetera and sometimes that's called 4e cognition and those are all like in allegiance together but that's what inactive points towards and self-evidencing is describing that the imperative of systems as modeled under active inference is for them to accrue consistent observations and find evidence for the kind of thing that they expect and prefer themselves to be in the coming weeks we'll explore this like expect and prefer and how can expectations be preferences and what about when we expect bad things to happen and uncertainty about expectations but we can contrast this imperative for finding consistent self-consistent or self-evidencing rather than pursuing rewarding so one can say i expect and prefer to be at room temperature and then moving around the space or putting the jacket on or off to reduce surprise about what observations one is receiving as opposed to needing to construct a reward framework in which room temperature is rewarding and then it gets less rewarding and then what is being maximized is the reward of the cognitive system and again in contrast self-evidencing puts the imperative on the system's self-model and reducing surprise about its own rather than seeing those observations as just proxy of reward states that have to be aaron uh i found both of those answers really interesting and i have a question to each of those so on the self evidencing we have so maybe this is fast forwarding a little bit in the book but they talk about both generative models and generative processes and those are distinct ideas so when we're talking about an organism as a model in its niche it's still distinct from the generative process right so that's something that i'm not 100 sure on and i

also find the self-evidencing part a little bit circular and i don't quite have a good understanding of that um in my mind the gender model and the gender process are very distinct and yes you're looking for you're an organism and you're in your niche and you want to succeed and by natural selection or whatever we only really see organisms that succeed they're going to look for things that meet their priors such as you know a nice homeostasis food reproduction all of that sort of stuff that's sort of my understanding of that part but it's still an organism using a model to navigate the world rather than it is a model itself um and then also on the idea of the minimizing free energy but also you know it's not a reward to me it sort of is a reward it's just a reward with respect to a prior distribution yes you're minimizing free energy and it's not and you know you have a prior preference of being at room temperature that's great but you're still you still have an object what an objective function or functional i think as the book calls it you're just subtracting the prior distribution of temperatures that your body likes so i still see that as like it's fundamentally an optimization problem it's just the units are a bit different and it's all statistical distributions and the differences between them rather than i don't know x minus $\bar{x}$ squared that sort of thing um so those are my those are the areas i'm not very clear on um so i don't know if anybody could clear those up so can i follow up on that then so on the inactivism i think the the problem is knowing what the specific meaning of it is compared with embodied and embedded because those those e's are so often used together what is where would you say an activism rather embodied and i i think you touched on it by saying about it involving him having a model maybe whereas being embodied i would suggest that inactive calls the action into focused whereas the embodiment in the embedded refer to just the materiality but the reason why they are allegianced as 4e eeee etc they're just fingers pointing at the moon they're just fingers pointing at real embodied organisms and some of the complexities of modeling them and framing them in contrast to disembodied approaches they're pointing at organisms and they're different theoretical vectors that have been taken and some of the synthesis amongst them has already been carried out yeah the an activism was became popularized the notion with the francisco varela by francisco varela actually and i think he was not he was not too fond of the term actually but anyway he the thing is that you wanted to stress the fact that cognition and consciousness emerges in the circularity sensation and action that was uh it was enacted in the circularity continuous circularity action and sensation so also feel it's really close to the free energy approach and active imprint and so although there are some early uh champions of an activism like um evan thompson and and some other scholars that argue that there are

actually crucial differences but anyway for what concerns the the the difference between the generative process and the generative model the generative process is basically the actual physical processes that turns causes into stimuli into that that impinge on our sensory surfaces the generative model is the best guess that an organism can make about that so they're not identical of course the the organism tries to learn at its best the generative process but you know it's just a model yes and i also wanted to mention um jacob howie's well-known paper the self-evidence in the brain which i believe is the seminal paper which introduced this concept of self-evidencing to active inference literature and uh in that paper he also contrasts this term with inactivism in activists and all the other four e-cognition frameworks and theories but uh briefly he believes that the self-evidencing brain by putting a boundary namely the markup blanket as one of its essential components in order for it to be able to infer the states of the environment it can retain its cognition within its inner statistical or generative model that's one of the reason uh the self-evidencing concept was deemed necessary to introduce at the first place and self-evidencing uh is something that just uh is beyond uh beyond the biological brain and it it can refer more generally and more broadly to any agent tries to learn about its environment as a one dynamical system which tries to perform inference about the other dynamical system which is the outside world but with a particular aim of trying towards the minimizing the error or performing the allostasis so to speak so that's one of the main differences between an activism and self-evidence in concept here it's also a very thank you um all for sharing it's very illustrative to look at figure 1.2 and see where these terms are used in the context of the low road and the high road to active inference which is one of the key concepts in this chapter and it's describing how the following chapters are going to be structured so we see self-evidencing on this upper branch of the subway system coming down from the high road and we see some of these other terms that people have been mentioning like generative model coming up from the low road and then in the um in the same page that we were just discussing i copied some sections from the text that are describing these high and low road they described that the high road is starting from the question of how living systems persist and act adaptively in the world and so that is very resonant with self-organization which is the first stop on that subway system from the high and also um varela's work autopoiesis a lot of questions about how cognizant sentient systems persist the high road perspective perspective is useful to understand what living organisms must do and why why to be persistent via surprise minimization not via reward maximization what they have to do so it's the what and the why in response to the broader question of how complex systems persist

that's the high road and we've had many discussions and fun for every person to also think about this high road low road distinction but the high road is starting from a complex system thriving or persisting in a complex niche we can think of and then the low road is starting from bayes theorem and continues up through the bayesian also motivating active inference from a different direction and it is describing the low road perspective is useful to illustrate how active inference agents minimize their free energy well they're not necessarily doing frequentist statistics they're doing something that we can describe as bayesian this textbook is going to introduce bayes theorem briefly in a box and it's a big area and some people will be familiar with thinking about priors and updating priors and the ways that that's discussed in bayesian statistics some will be familiar for some it might be novel but this is describing kind of the atomic computational or statistical uh nucleus that is going to describe how different systems can fulfill that imperative that we saw traced out by the high road and ali i know you'll have a lot to add on uh i've also put a pdf to carver stream's essay beyond the desert landscape on resources and resources page and also on the chapter one uh questions and notes page uh i believe this essay is the most comprehensively articulated and in my opinion very beautifully even poetically articulated account of the high road and lower distinctions and the philosophical distinctions between them uh if you scroll down i think you can find the paper i've attached there yeah right at the bottom of the page and i've also inclu appended yeah that's it the carl for instance no below that yeah beyond the desert landscape uh and i've also appended uh andy clark's reply to this essay because uh this chapter comes from this book called andy clark and its critics which uh i think can uh help to understand uh the real differences between those two um concepts and why it's important to distinguish between them when we want to discuss and understand any active inference related concepts thank you awesome thanks i'm gonna just make note of that in here maybe just a comment on the why if you daniel uh like talking about the the high road is it is it why really because my understanding really is uh the perspective or the nature of the free energy principle is really a scar says is more tautological than teleological and why seems to point to some sort of teleology so it's go ahead go ahead yes scientific or formal addressments of teleology or end or goal directedness are very interesting and another framework or theory which is evolution has also threaded the needle with tautology and teleology and so there's a lot to be said there um about how those two categories aren't necessarily disjoint do you believe they have to be disjoint or are you expecting or preferring that they are um if the if the imperative is to persist and survive and we simply don't observe systems that don't that phenomena can be both taught

logical in the sense that it's self-describing as well as teleological in that from a perspective's view not speaking for the system itself but from our view on the system it is acting as if it is trying to survive the bubble is acting as if it's trying to get to the surface of the water in the sense that it's not acting like it's trying to go deeper we don't need to entail any qualia or any sort of broader teleological reason for that to occur we can describe something that is both teleological and tautological yes well i i think i think i agree i think that from what what car or often says that it's you know deflationary accounts is that basically the free energy principle basically basically describes self-organizing system uh just because it persists and so what characteristics does it need in order to persist and so in in that sense is uh topological so it's just uh you know if it persists in it must behave like that you don't even need to put in any imperative to survive if it survives it must behave like that but of course as you said you can frame this self-organizing drive in a sense as some sort of imperative to to maintain its form and function of the organism so there's a lot of interesting things to say about that and uh also one note is sometimes we speak in a mode of like what the imperative is for the system as itself in terms of what it normatively should or imperatively must be doing but also we can just speak from ourselves as modelers of other things as what our modeling imperatives are and that is a slightly different and that's sometimes known as instrumentalism which can be compatible with realism of various flavors and it's been explored broadly however from our perspective as observers things that are not acting as to minimize their surprise relative to some area in space that makes us say it's that kind of a thing if it's not staying in that space which we're defining as that kind of thing it's not that thing if the tornado were not staying in some range of phase space that we call tornado it wouldn't be a tornado to us and so sometimes that can be a more empirically grounded and less speculative stance to have two feet as us as modeling systems which is going to be um consistent with the meta bayesian approach that's discussed later in the book and it sidesteps or it preempts a lot of very thorny questions around speaking for what systems like actually are or actually are doing and it's the difference between saying this the the thing is a model of its niche you know big if true we're going to model it as if it's a model of its niche it's irrefutable that's just a choice that we're taking and someone could say well i would have taken a different choice in modeling and you can say great what would your choice have been and there's actually a starting point instead of just this endless speculation around what is the system really and especially when we think about our markov blankets and our inability to make direct contact to hidden states it it bolsters the the justification of speaking as a modeler very cool um anybody else with

comments on this uh question this self-evidencing and an activism which we could talk i'm sure all day about so if we just finish off that i think possibly this it's the language that's the problem with self-evidencing it just sounds as though it's self confirmatory all it's doing is looking for support bricks existence which is not a good strategy for survival uh why you say that well if i only look for evidence of my existence rather than things that are a challenge to that it i mean is possibly related to the dark room problem yes um exactly yeah in the for the sake of long-term uh persistence it may be necessary to incur sometimes transiently into states that are surprising so there is no real there's no real uh incoherence there i think no i i agree i'm just uh challenging it it's as a language a bit of a language barrier yeah great great point and indeed the natural language is often heavily shading interpretations and people's conceptions of different terms like around surprise belief expectation preference all these everyday terms which are core in the act of inference ontology it's really important to interrogate our understanding of them and check their alignment with formalisms and um this book is going to a large extent focus on the kernel of the active inference action perception loop single agent single level and in that way it's very analogous to like a single linear regression now in practice linear aggression on large data sets does sweeps across families of models and there's all kinds of things that come into play just for the linear aggression that's the kernel and so in a nested model one could reduce surprise and find confirmatory evidence i'm the kind of thing that seeks out novel information sources therefore seeking out novel information sources is confirmatory evidence of that so once we step into multi-layer models many of the absolutely valid limitations of the kernel in and of itself are addressed so those are like all the directions that people's questions on and written down are super important they're all the things that that are direct adjacencies and relevant for applications and there's going to be things about real complex systems that the kernel does not describe totally so this isn't like the end of the active inference modeling of complex systems total book it really is just describing like the essence of the loop so i want to just give aaron the last um comment because it's the top of the hour now so yeah i just wanted to add to neil's question again like i fully understand where you're coming from neil uh when i first read about the self-evidencing thing i was like that makes zero sense and exactly what you described like oh okay you're not gonna look for predators like as a basic thing that's not evidence for my existence that's evidence for my death um but i think the way i've sort of come around to it is it's all about the prior preferences and what you expect to see um and yeah these this occupation of face space i think is quite important of you you sort of get attracted to particular states where you're comfortable but you do need to know

about the rest of the world anyway and you have to plan for not being in those states sometimes um and so looking for evidence of your existence is all about matching those priors and those priors can be pretty flexible um so yeah that's i think that will in my mind like that comes through a bit later on in the book i think so i would say my advice do is stick with it um but it is it was very counter-intuitive to me as well when i first came across it